

4.2.3 Orchard Key Components

Let $\ell_{\text{PRFexpand}}$, ℓ_{sk} , ℓ_{ovk} , ℓ_{d} , and ℓ_{dk} be as defined in § 5.3 ‘Constants’ on p. 77.

Let $\mathbb{P}$, $\text{repr}_{\mathbb{P}}$, $\ell_{\mathbb{P}}$, $q_{\mathbb{P}}$, and $r_{\mathbb{P}}$ be as defined in § 5.4.9.6 ‘Pallas and Vesta’ on p. 108.

Let $\text{Extract}_{\mathbb{P}}$ be as defined in § 5.4.9.7 ‘Coordinate Extractor for Pallas’ on p. 109.

Let $\text{GroupHash}_{\mathbb{P}}$ be as defined in § 5.4.9.8 ‘Group Hash into Pallas and Vesta’ on p. 110.

Let $\text{PRF}^{\text{expand}}$ and $\text{PRF}^{\text{ockOrchard}}$ be as defined in § 5.4.2 ‘Pseudo Random Functions’ on p. 89.

Let $\text{DeriveInternalFVK}^{\text{Orchard}}$ be as defined in [ZIP-32, Orchard internal key derivation].

Let $\text{PRP}^{\text{d}} : \mathbb{B}^{\ell_{\text{dk}}/8} \times \mathbb{B}^{\ell_{\text{d}}} \rightarrow \mathbb{B}^{\ell_{\text{d}}}$ be as defined in § 5.4.4 ‘Pseudo Random Permutations’ on p. 91.

Let $\text{KA}^{\text{Orchard}}$, instantiated in § 5.4.5.5 ‘Orchard Key Agreement’ on p. 93, be a key agreement scheme.

Let $\text{Commit}^{\text{ivk}}$, instantiated in § 5.4.8.4 ‘Sinsemilla commitments’ on p. 101, be a commitment scheme.

Let $\text{DiversifyHash}^{\text{Orchard}}$ be as defined in § 5.4.1.6 ‘DiversifyHash^{Sapling} and DiversifyHash^{Orchard} Hash Functions’ on p. 81.

Let $\text{SpendAuthSig}^{\text{Orchard}}$ instantiated in § 5.4.7.1 ‘Spend Authorization Signature (Sapling and Orchard)’ on p. 98 be a signature scheme with re-randomizable keys.

Let I2LEBSP , I2LEOSP , and LEOS2IP be as defined in § 5.1 ‘Integers, Bit Sequences, and Endianness’ on p. 76.

Define $\text{ToBase}^{\text{Orchard}}(x : \mathbb{B}^{\ell_{\text{PRFexpand}}/8}) := \text{LEOS2IP}_{\ell_{\text{PRFexpand}}}(x) \pmod{q_{\mathbb{P}}}$.

Define $\text{ToScalar}^{\text{Orchard}}(x : \mathbb{B}^{\ell_{\text{PRFexpand}}/8}) := \text{LEOS2IP}_{\ell_{\text{PRFexpand}}}(x) \pmod{r_{\mathbb{P}}}$.

Define $\text{DeriveDkAndOvk}^{\text{Orchard}}(\text{rivk} : \mathbb{F}_{r_{\mathbb{P}}}, \text{ak} : \{0 \dots q_{\mathbb{P}} - 1\}, \text{nk} : \mathbb{F}_{q_{\mathbb{P}}})$ as follows:

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let  $K = \text{I2LEBSP}_{\ell_{\text{sk}}}(\text{rivk})$ 
let  $R = \text{PRF}_K^{\text{expand}}([0\text{x}82] \parallel \text{I2LEOSP}_{256}(\text{ak}) \parallel \text{I2LEOSP}_{256}(\text{nk}))$ 
let  $\text{dk} : \mathbb{B}^{\ell_{\text{dk}}/8}$  be the first  $\ell_{\text{dk}}/8$  bytes of  $R$  and let  $\text{ovk} : \mathbb{B}^{\ell_{\text{ovk}}/8}$  be the remaining  $\ell_{\text{ovk}}/8$  bytes of  $R$ .
return  $(\text{dk}, \text{ovk})$ 

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A new Orchard *spending* key sk is generated by choosing a *bit* sequence uniformly at random from $\mathbb{B}^{\ell_{\text{sk}}}$.

From this *spending* key, the *Spend authorizing* key $\text{ask} : \mathbb{F}_{r_{\mathbb{P}}}^*$, the *Spend validating* key $\text{ak} : \{0 \dots q_{\mathbb{P}} - 1\}$, the *nullifier deriving* key $\text{nk} : \mathbb{F}_{q_{\mathbb{P}}}$, the $\text{Commit}^{\text{ivk}}$ randomness $\text{rivk} : \mathbb{F}_{r_{\mathbb{P}}}$, the *diversifier* key $\text{dk} : \mathbb{B}^{\ell_{\text{dk}}/8}$, the $\text{KA}^{\text{Orchard}}$ private key $\text{ivk} : \{1 \dots q_{\mathbb{P}} - 1\}$, the *outgoing viewing* key $\text{ovk} : \mathbb{B}^{\ell_{\text{ovk}}/8}$, and corresponding “internal” keys are derived as follows:

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let mutable  $\text{ask} \leftarrow \text{ToScalar}^{\text{Orchard}}(\text{PRF}_{\text{sk}}^{\text{expand}}([6]))$ 
let  $\text{nk} = \text{ToBase}^{\text{Orchard}}(\text{PRF}_{\text{sk}}^{\text{expand}}([7]))$ 
let  $\text{rivk} = \text{ToScalar}^{\text{Orchard}}(\text{PRF}_{\text{sk}}^{\text{expand}}([8]))$ 
if  $\text{ask} = 0$ , discard this key and repeat with a new  $\text{sk}$ .
let  $\text{ak}^{\mathbb{P}} = \text{SpendAuthSig}^{\text{Orchard}}.\text{DerivePublic}(\text{ask})$ 
if the last bit (that is, the  $\tilde{y}$  bit) of  $\text{repr}_{\mathbb{P}}(\text{ak}^{\mathbb{P}})$  is 1:
    set  $\text{ask} \leftarrow -\text{ask}$ 
let  $\text{ak} = \text{Extract}_{\mathbb{P}}(\text{ak}^{\mathbb{P}})$ 
let  $\text{ivk} = \text{Commit}_{\text{rivk}}^{\text{ivk}}(\text{ak}, \text{nk})$ 
if  $\text{ivk} \in \{0, \perp\}$ , discard this key and repeat with a new  $\text{sk}$ .
let  $(\text{dk}, \text{ovk}) = \text{DeriveDkAndOvk}^{\text{Orchard}}(\text{rivk}, \text{ak}, \text{nk})$ 
let  $(\text{ak}_{\text{internal}}, \text{nk}_{\text{internal}}, \text{rivk}_{\text{internal}}) = \text{DeriveInternalFVK}^{\text{Orchard}}(\text{ak}, \text{nk}, \text{rivk})$ 

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